

6-1 Powers and Exponents

Music Studio | Standard: 6.EE.1 | Color-by-Number Practice

Learning Goal: I can write and evaluate numbers in exponent form using a base and a power.

Directions: Solve each problem. Match each answer to the Color Code. On the next page, color every box with that problem number using the matching color.

Vocabulary: base, exponent, power, evaluate

Explain one: I solved problem ___ by ___. Because ___, my answer is ___.

Solve First

1. Evaluate 2^3 . Work: _____ Answer: _____	2. Evaluate 3^2 . Work: _____ Answer: _____
3. Evaluate 4^2 . Work: _____ Answer: _____	4. Evaluate 5^2 . Work: _____ Answer: _____
5. Evaluate 2^4 . Work: _____ Answer: _____	6. Evaluate 10^2 . Work: _____ Answer: _____
7. Evaluate 6^2 . Work: _____ Answer: _____	8. Evaluate 3^3 . Work: _____ Answer: _____
9. Evaluate 2^5 . Work: _____ Answer: _____	10. Evaluate 7^2 . Work: _____ Answer: _____
11. Evaluate 4^3 . Work: _____ Answer: _____	12. Evaluate 5^3 . Work: _____ Answer: _____

Quick check: Before coloring, make sure every answer appears in the Color Code on the next page.

6-1 Mystery Picture: Beat Board Mosaic

Solve a problem, match the answer to a color, then color every box with that problem number.

Color Code	Color Code	Color Code	Color Code
8 = red	9 = orange	16 = yellow	25 = green
100 = blue	36 = purple	27 = pink	32 = brown
49 = gray	64 = teal	125 = black	

2	2	1	2	2	2	3	8	3	4	4	6
2	5	5	5	5	5	6	6	6	7	9	6
1	5	8	9	8	8	9	9	9	12	11	6
3	6	10	2	2	1	2	2	4	2	11	6
3	6	9	2	5	5	5	7	5	2	9	6
3	6	9	1	5	11	11	7	4	12	9	6
5	8	11	3	6	12	1	10	4	2	11	8
5	8	11	3	8	8	10	8	5	2	11	8
5	8	11	5	5	5	4	5	5	4	11	11
7	10	3	4	4	2	3	3	4	4	2	10
7	12	12	1	11	11	12	12	12	2	2	11
9	9	9	8	8	8	9	9	9	10	11	11

Early finisher writing: Choose one problem and explain your thinking in one complete sentence: Because ___, I ___.

Teacher Answer Key

#	Answer	Color	Teacher Tip
1	8	red	Multiply the base by itself the exponent number of times.
2	9	orange	Multiply the base by itself the exponent number of times.
3	16	yellow	Multiply the base by itself the exponent number of times.
4	25	green	Multiply the base by itself the exponent number of times.
5	16	yellow	Multiply the base by itself the exponent number of times.
6	100	blue	Multiply the base by itself the exponent number of times.
7	36	purple	Multiply the base by itself the exponent number of times.
8	27	pink	Multiply the base by itself the exponent number of times.
9	32	brown	Multiply the base by itself the exponent number of times.
10	49	gray	Multiply the base by itself the exponent number of times.
11	64	teal	Multiply the base by itself the exponent number of times.
12	125	black	Multiply the base by itself the exponent number of times.

Use note: This file is original practice aligned to the lesson skill. It is designed for low-prep review, centers, sub plans, or early finishers.